Evaluating and Comparing Aerodynamic Design Efficiency Throughout Flight History via Lift and Drag Analysis of Three Airfoils from Three Eras of Flight

University of California, Irvine | Samueli School of Engineering
Department of Mechanical and Aerospace Engineering

MAE 108: Aerospace Laboratory Spring 2024
Design Project Group Presentation

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Lab Group 3 (Tu. 2:00 - 4:50 pm)
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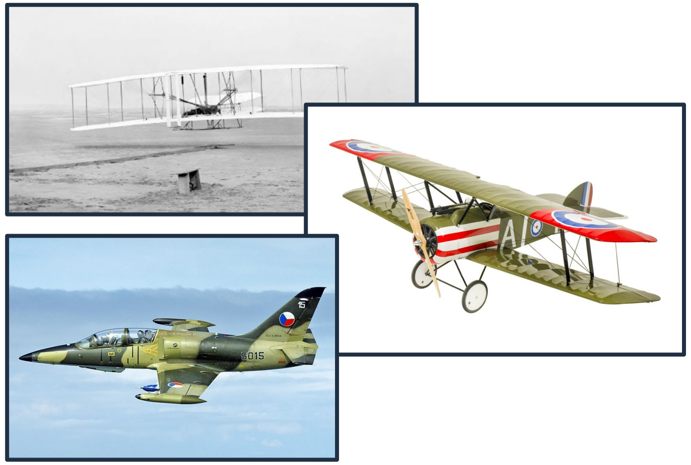
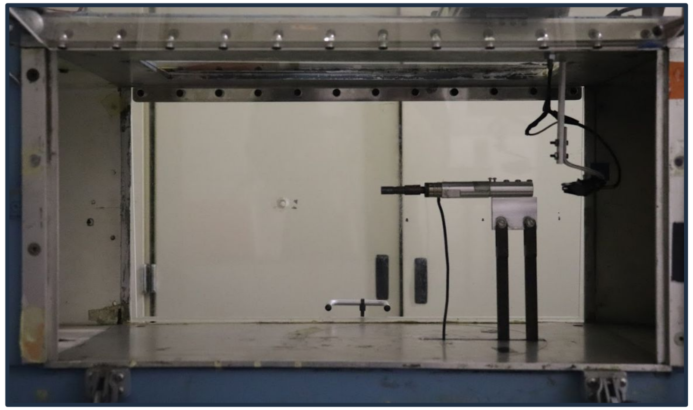
UCI Samueli School of Engineering
Department of Mechanical and Aerospace Engineering

Motivation



Exploring the last century of flight technology: How has humanity improved upon airfoil designs? What trends can we analyze moving forward?

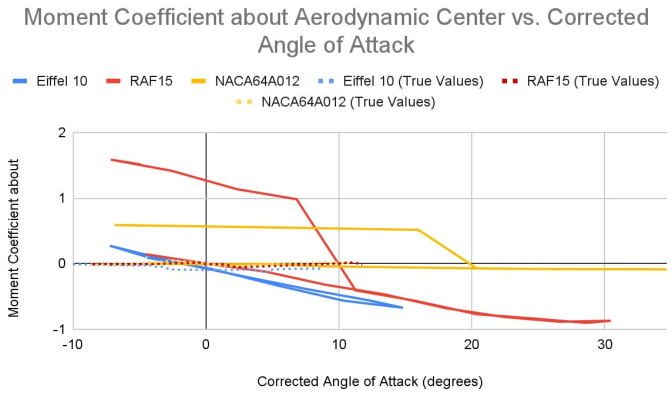
Method

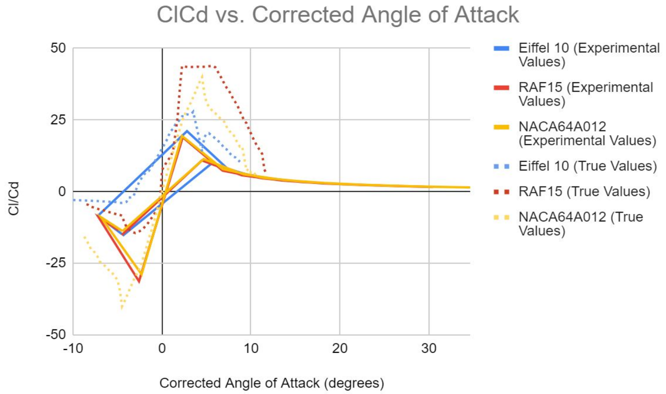


Lift-drag analysis via low-speed wind tunnel at 20 m/s for 3D printed airfoils:

1. Wright Brothers' Eiffel 10
2. Sopwith Camel's RAF15
3. L-39C's NACA 64A012

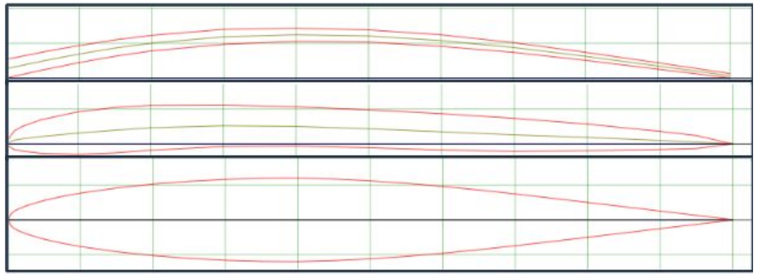
Results



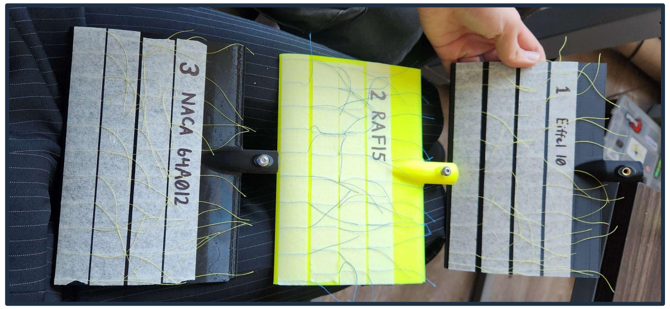


This talk presents an experimental analysis and comparison of three airfoils throughout the history of flight

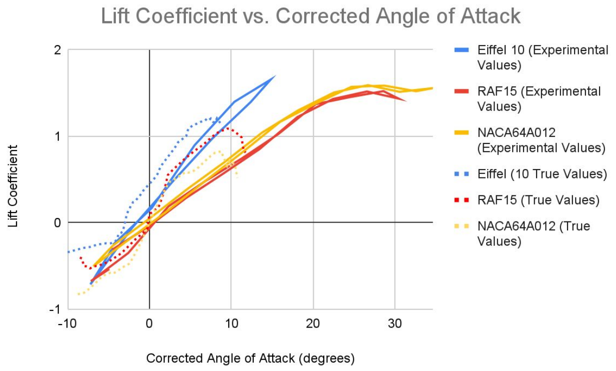
1. Objectives / Literature Review



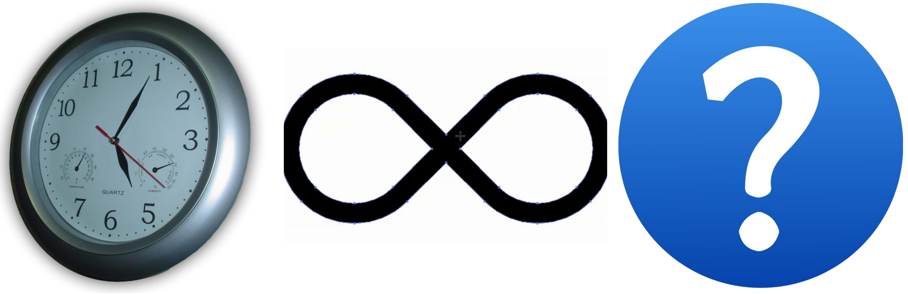
2. Methodology / Procedure



3. Experimental Results / Data



4. Conclusions / Future Considerations



This project aims to conduct a:

1. Comparative Analysis of Aerodynamic Efficiency

- Lift-to-Drag Ratio of all configurations
- Efficiency Analysis

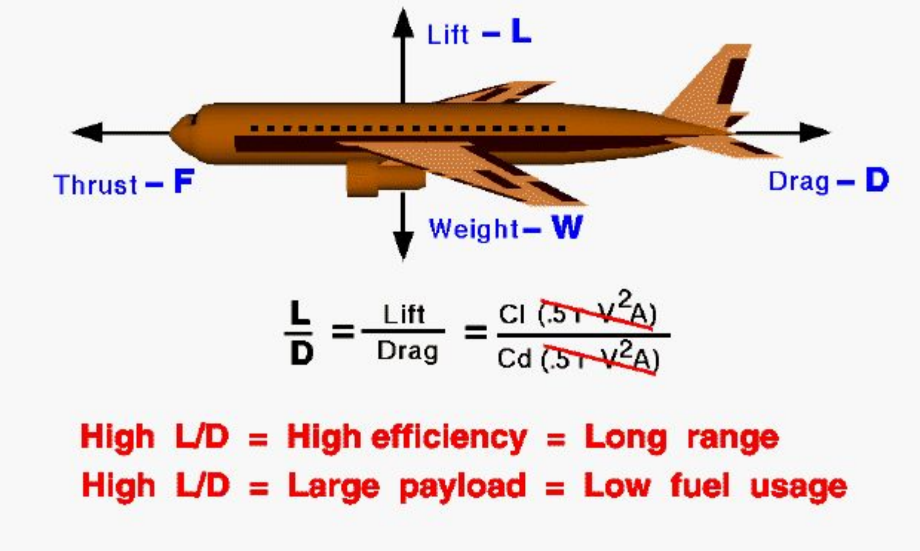


Figure [1]. Lift-to-Drag Ratio and its relation to efficiency

2. Experimental Validation and Visualization of Aerodynamic Performance

- Recording and analyzing quantitative performance data
- Flow visualization techniques

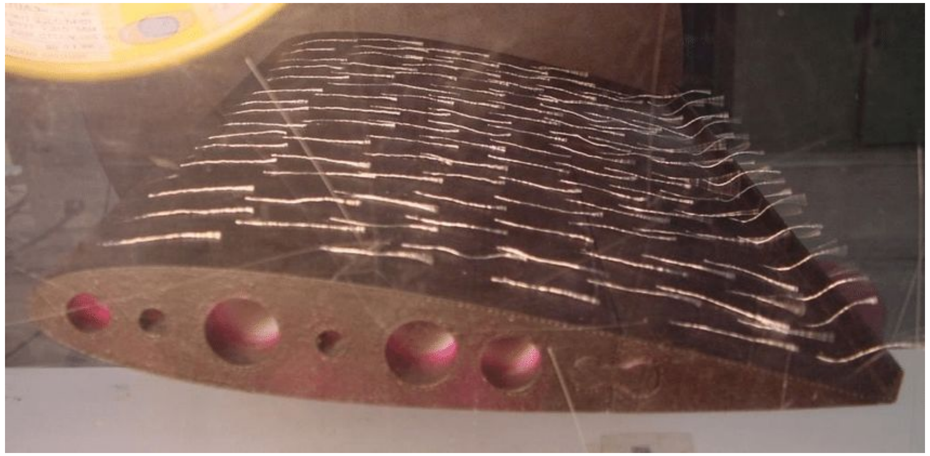


Figure [2]. Example of airfoil with tufts

Project Inspirations and Past Research

8 Sources used/referenced in determining experimental airfoils

1. What the Wright Brothers Did and Did Not Understand About Flight Mechanics—In Modern Terms (Culick)
2. Performance Analysis and Tactics of Fighter Aircraft from WWI (Eberhardt)
3. Sopwith Camel Versus Fokker Dr.1: Which one turns the best? An Analysis of the Stationary and Instantaneous Turn Performance for Two of WW1's Most Famous Airplanes (Jonsson)
4. Effect of the wing airfoil shape on the aerodynamics and performance of a jet-trainer aircraft – An optimization approach (K A Othman and A S Mahdi Al-Obaidi)
5. Study and analysis of lift to drag ratio performance of supercritical wing based on computational fluid dynamics method (Xuyang Bai 2023)
6. Comparison of Aerodynamic Performances of Various Airfoils from Different Airfoil Families Using CFD (Kaya, Kurt, Kök)NASA Supercritical Airfoils (Charles D. Harris)
7. Evolutionary generative design of supercritical airfoils: an automated approach driven by small data (Sun et. al.)

Project Inspirations and Past Research

1 - Wright Brothers and how they first analyzed flight, how that evolved, referenced to find the airfoil used

2 - WWI fighter plane analysis that includes 18 different models and their numerical data, serves as basis for understanding of technology during the time

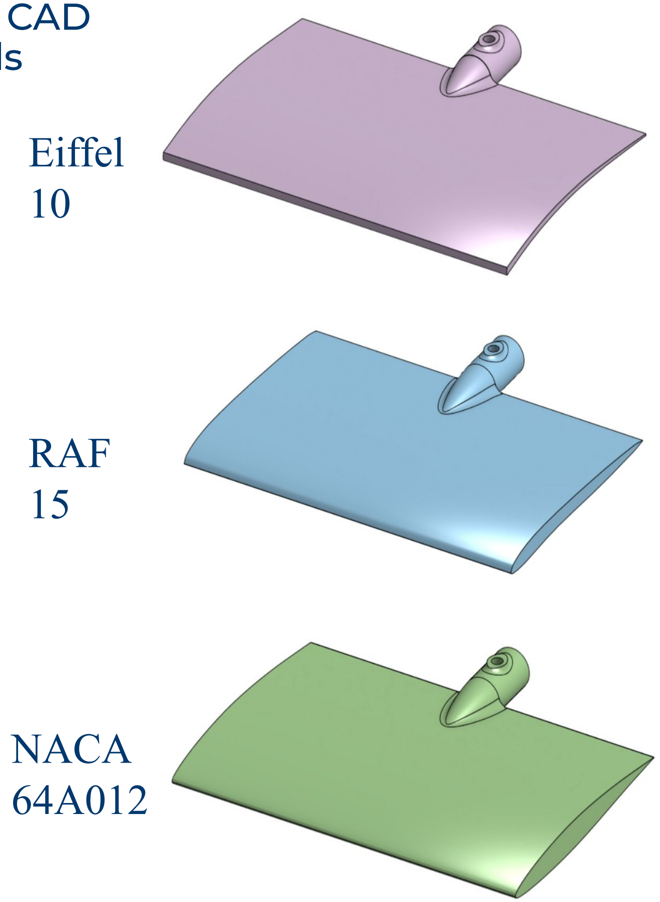
3 - Sopwith Camel vs. Fokker turning, analysis that focuses on 2 specific planes, used as source for understanding comparisons

4 - NACA 64A012 airfoil performance and comparison to other airfoils, used as modern example due to data provided on the plane and clear purpose/goals (fighter plane trainer)

Figure [3]- Sopwith Camel, uses RAF15 airfoil

Project Inspirations and Past Research

Figure [4]. - Airfoil CAD Models



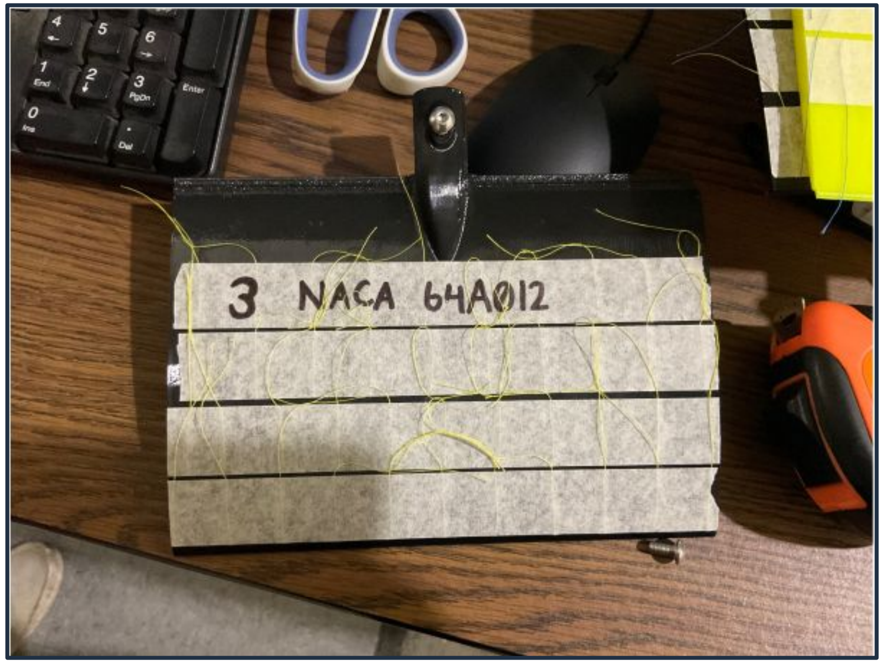
5 - Aerodynamic performance of airfoils using CFD, shows example for experiment utilizing different angles of attack

6 - Lift to drag ratio performance of airfoils analyzed, provide example for how we can think of ours

7 - Supercritical airfoil information history and reasoning behind how they are improved

8 - Supercritical airfoil generation and how that comes from the manual improvement process

Design Method: Airfoils



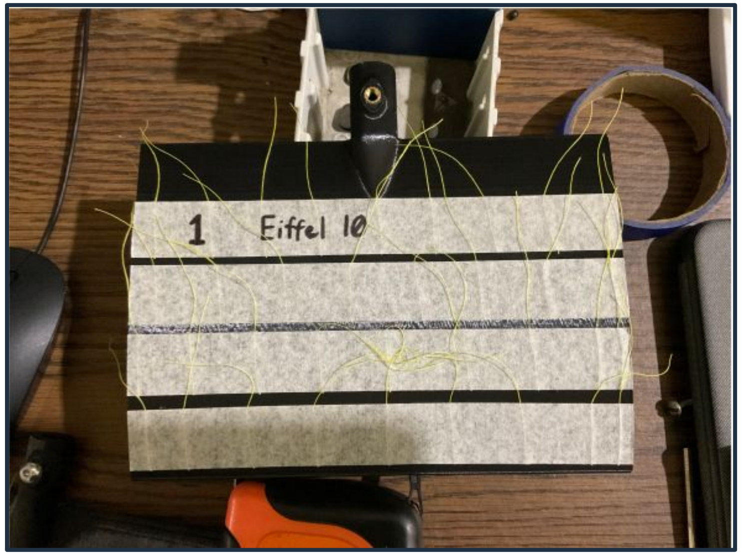


Figure [5]. - Airfoil models 3D printed

Methodology and Procedure

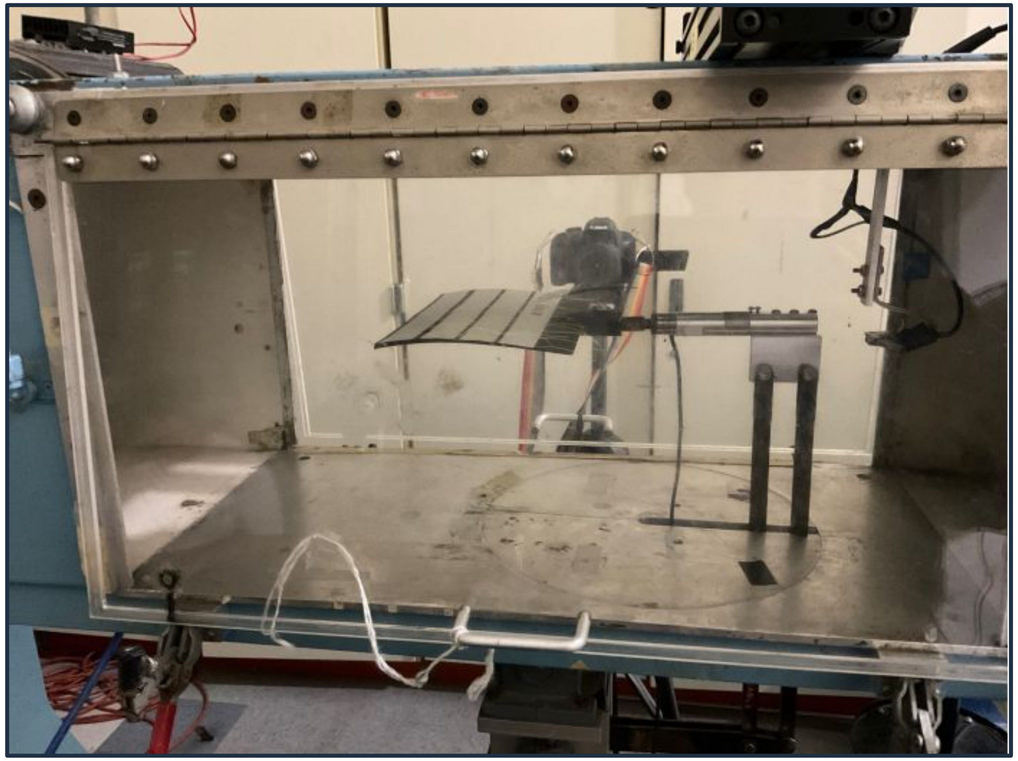


Figure [6]. - Wright Brothers' Eiffel 10 Print secured in test section (referred to as T.S.).

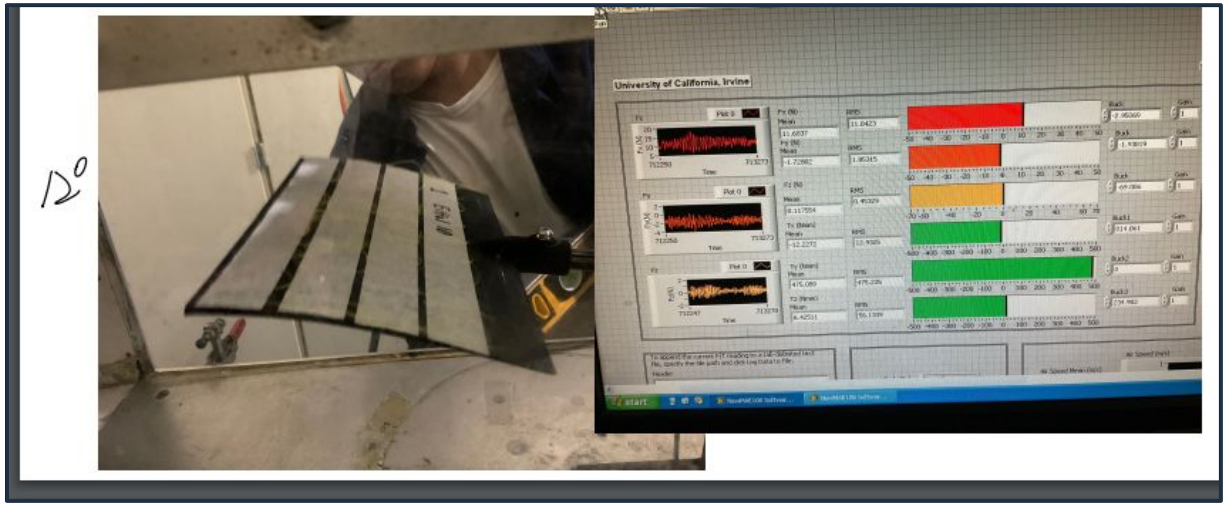


Figure [7]. - Eiffel 10 close/at stall and a measurement of the force data through LabView software.

The Airfoil is shaken hard and the airfoil floss tufts are completely separate support it been stalled.

Methodology and Procedure:

Figure [8]. - Wright Brothers' Eiffel 10 print at stall point.

Methodology and Procedure

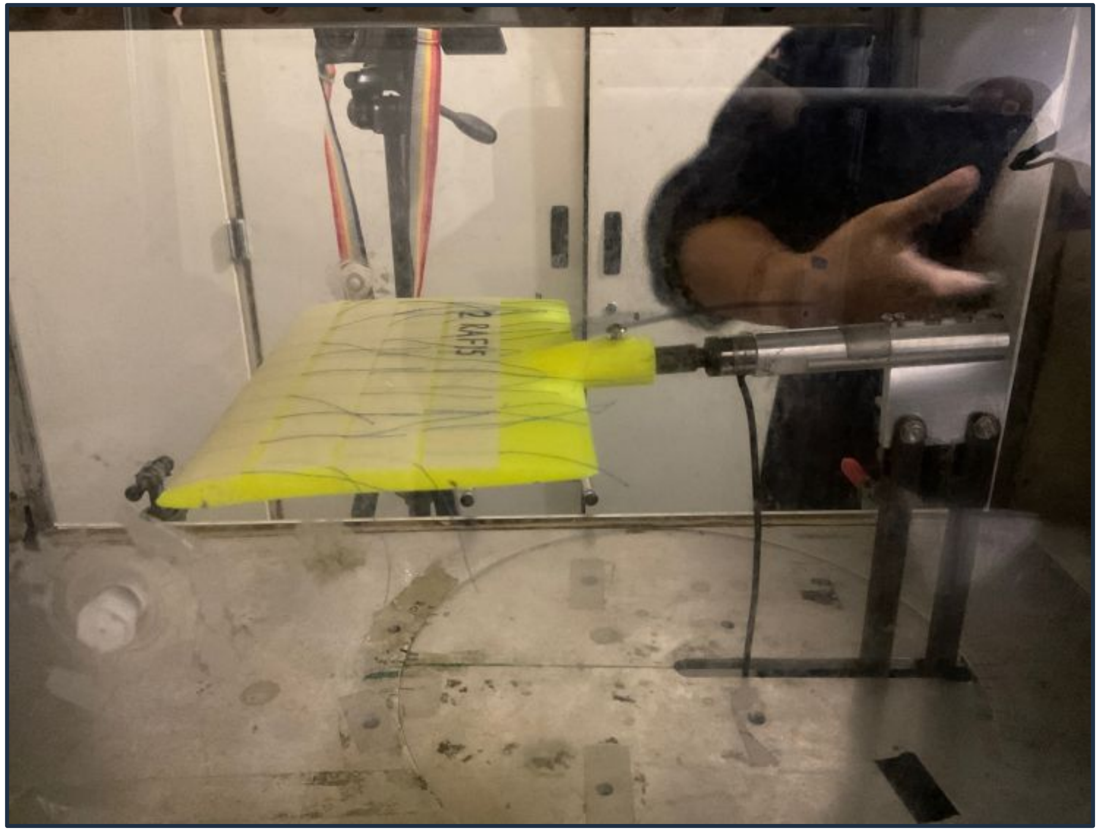


Figure [9]. - Camel RAF15 print within the T.S.

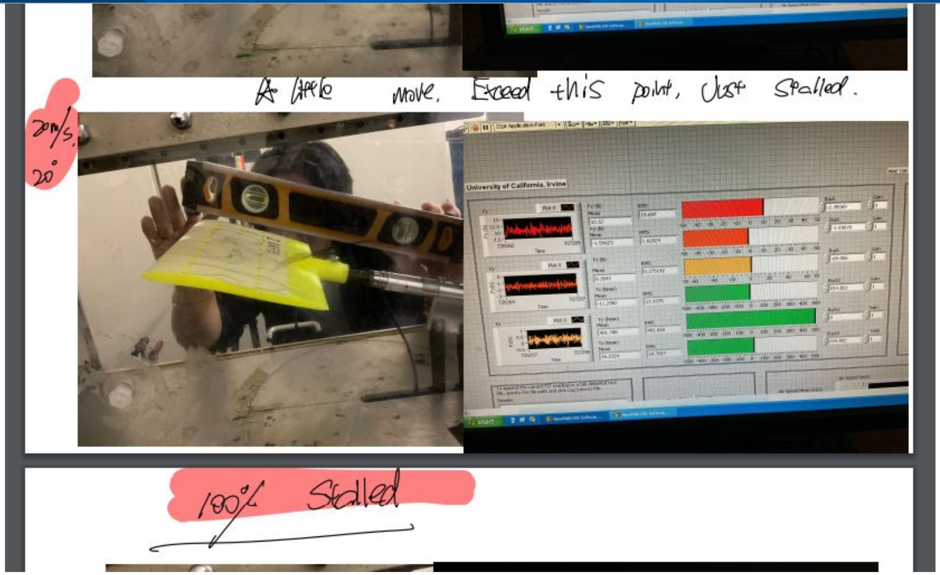


Figure [10]. - Camel RAF15 at stall with captured force data measurements through LabView

Around 17 degrees AOA, the airfoil is stalled briskly, which was shown by a dramatic change in the floss tufts.

Methodology and Procedure

Figure [11]. - NACA 64A012 print within the T.S.

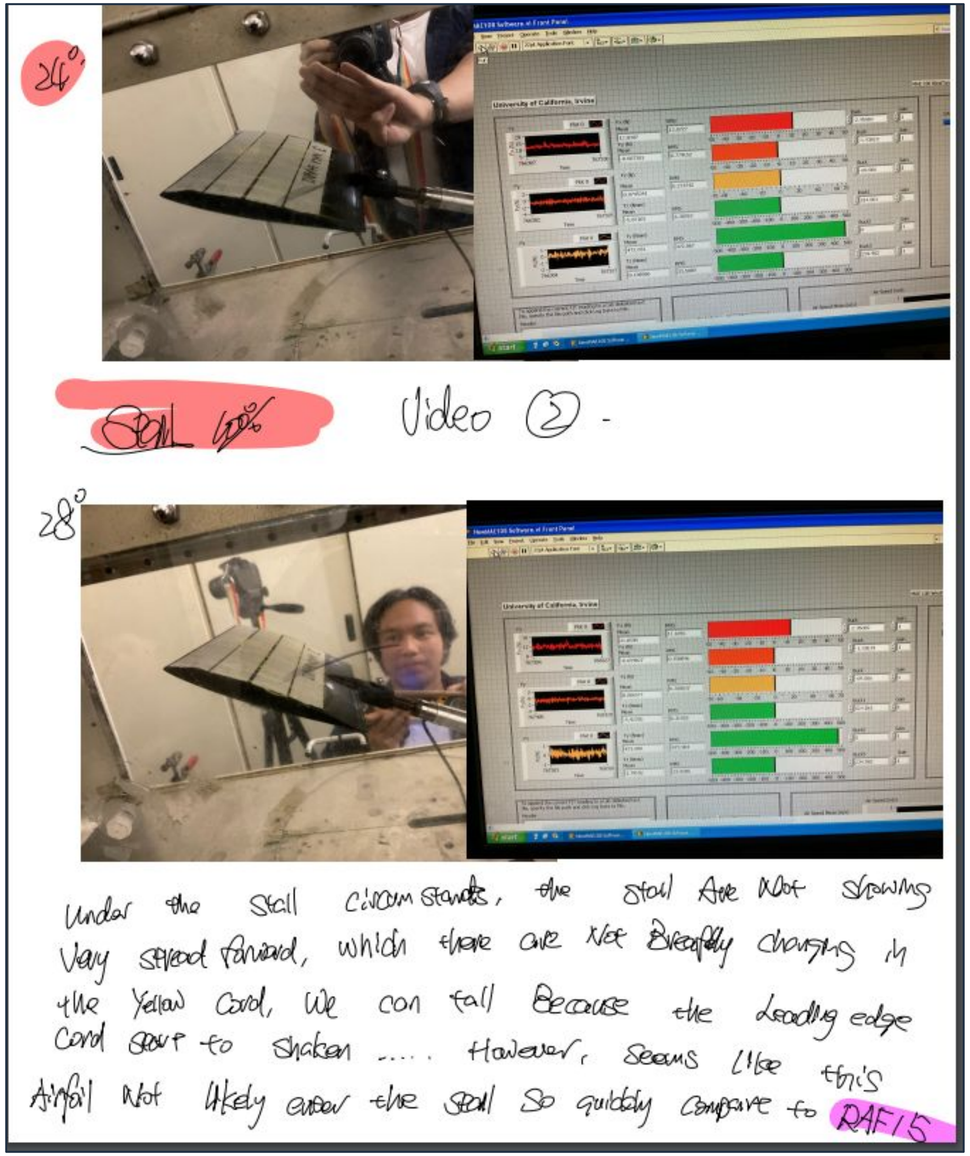


Figure [12]. - NACA 64A012 at stall along with LabView force measurements.

Methodology and Procedure

Figure [13]. - NACA 64A012 print at stall

Results

- Coefficient of lift has greater range over time
- Newer Airfoils have shallower curves for drag coefficient

Airfoil	Peak Lift (Newtons)	Peak Drag (Newtons)
Eiffel 10	11.40394227	2.544162985
RAF15	10.48777183	5.540561459
NACA64A012	10.96392977	6.770378671

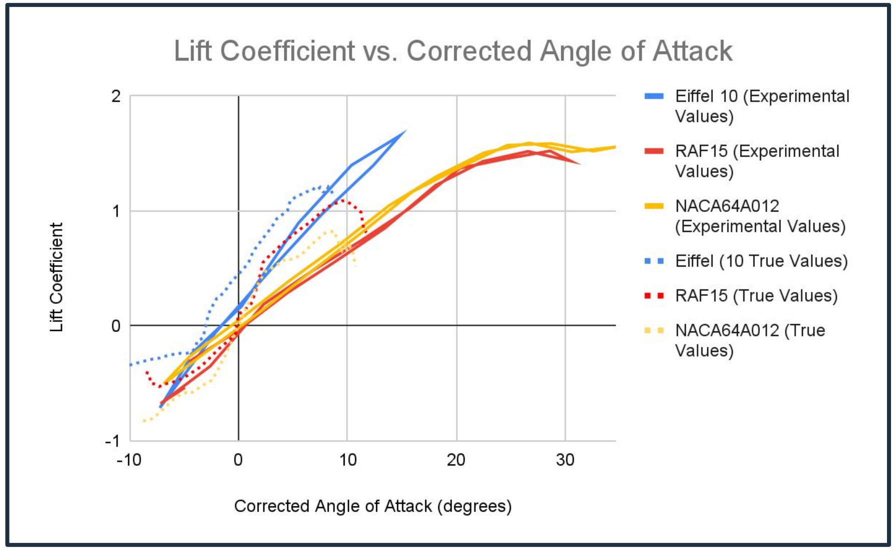


Figure [14]. - C_L vs. Corrected AoA w/ Real Data

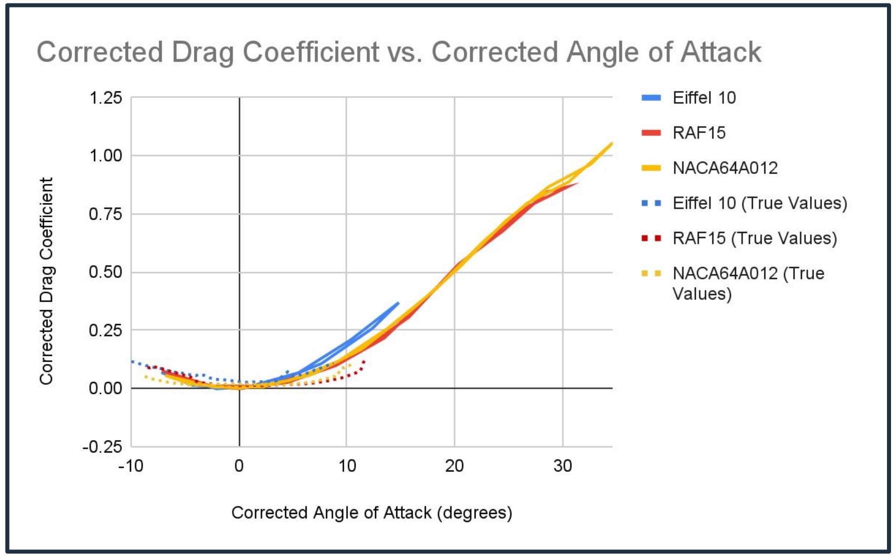


Figure [15]. - C_D vs. Corrected AoA w/ Real Data

Results

- Moment Coefficient trends towards a flatter curve
- C_L/C_D trends towards steeper curves

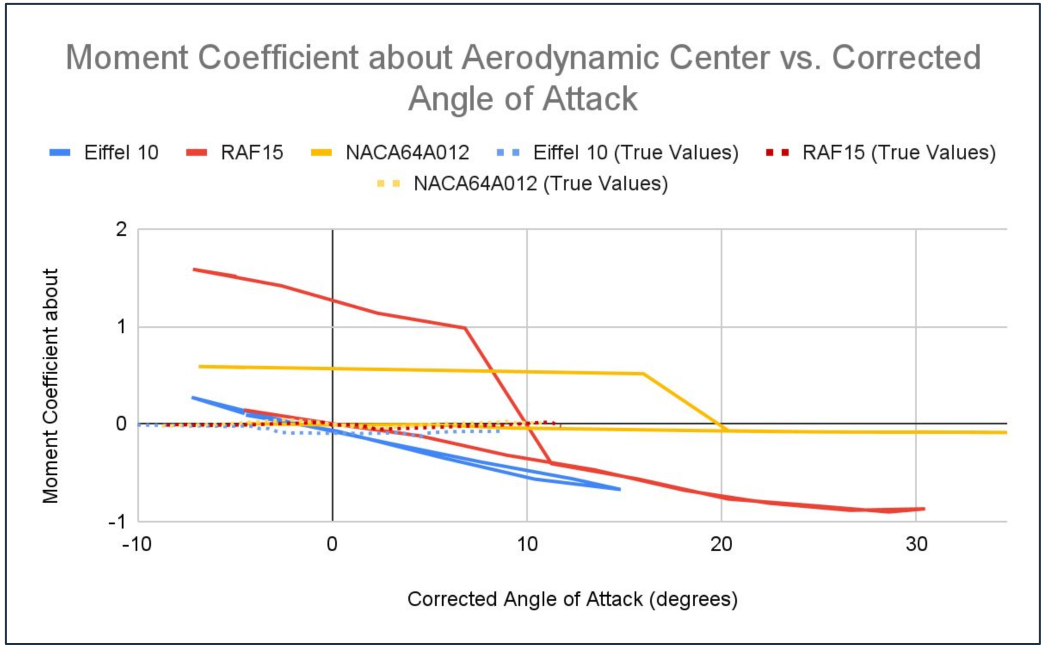


Figure [16]. - C_M vs. Corrected AoA w/ Real Data

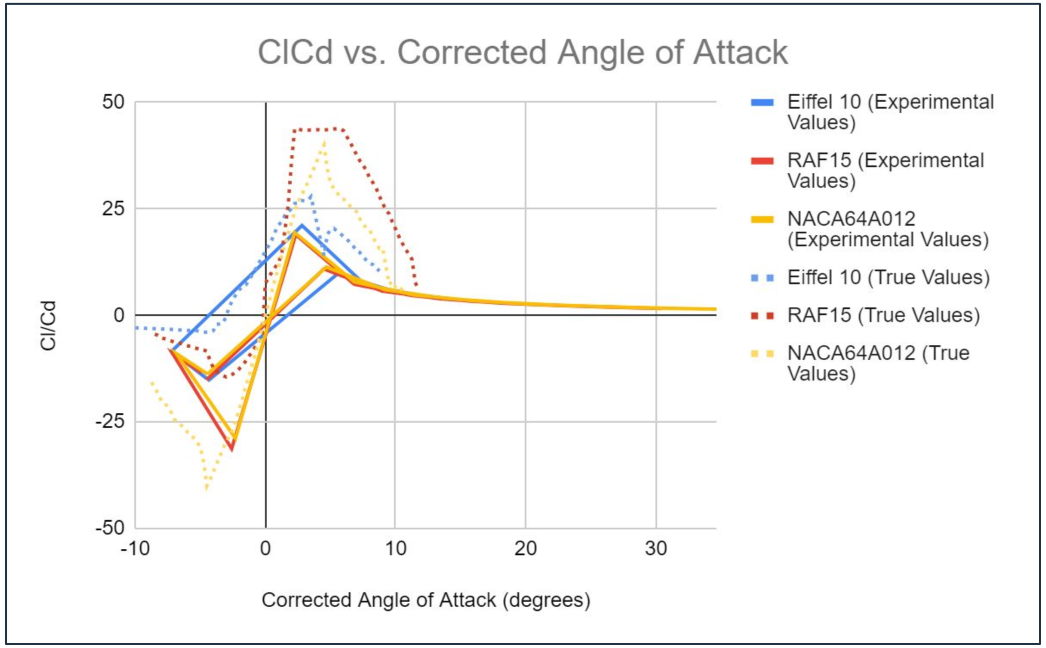


Figure [17]. - Efficiency Factor vs. Corrected AoA w/ Real Data

Error Analysis

Airfoil Model Generation

- 3D Printing Variations
- Airfoils were not all printed from the same source

Equipment Capability

- Cannot record negative sting values

Measurement Uncertainties

- Inflated values due to wind tunnel

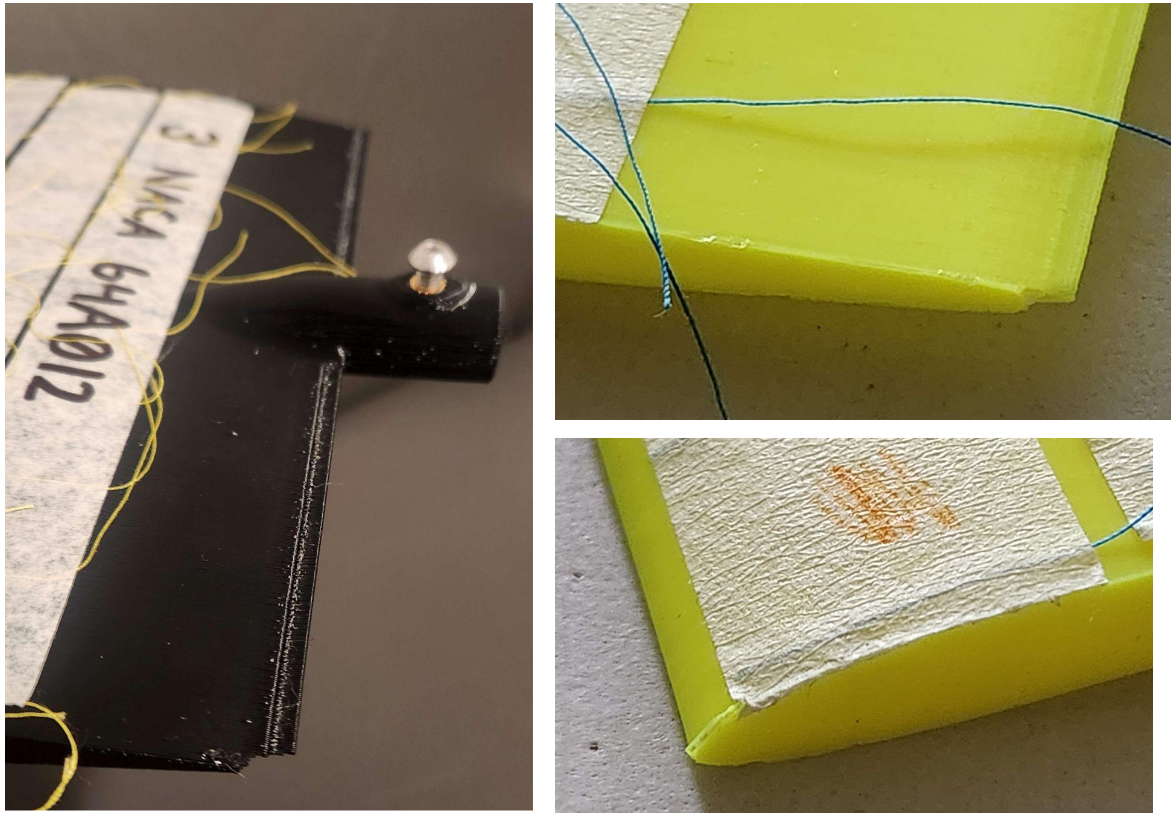
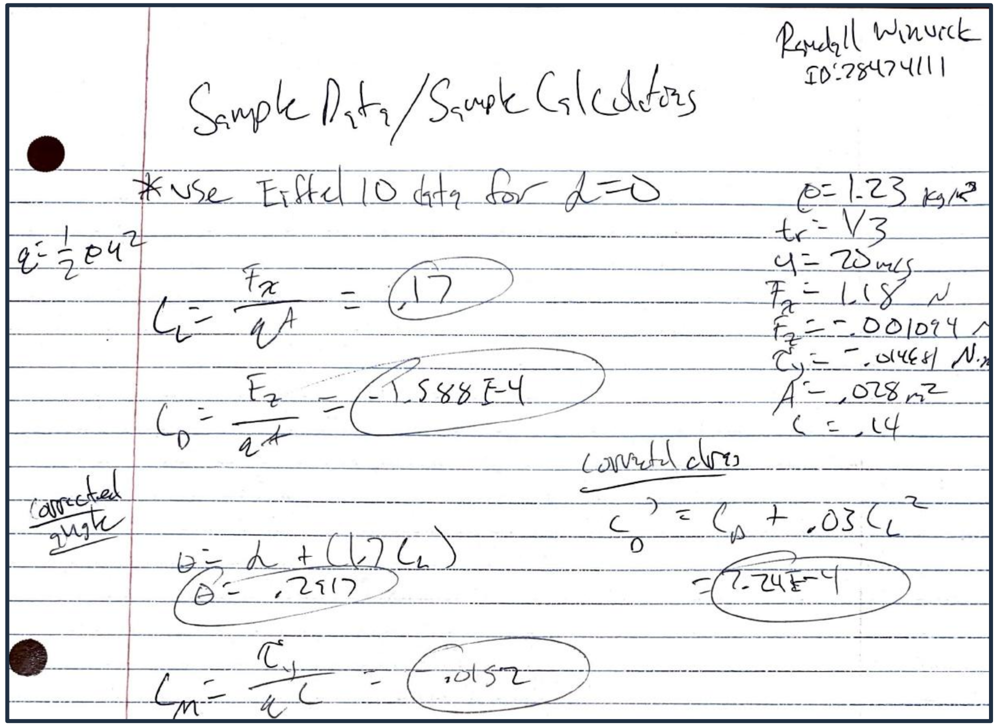
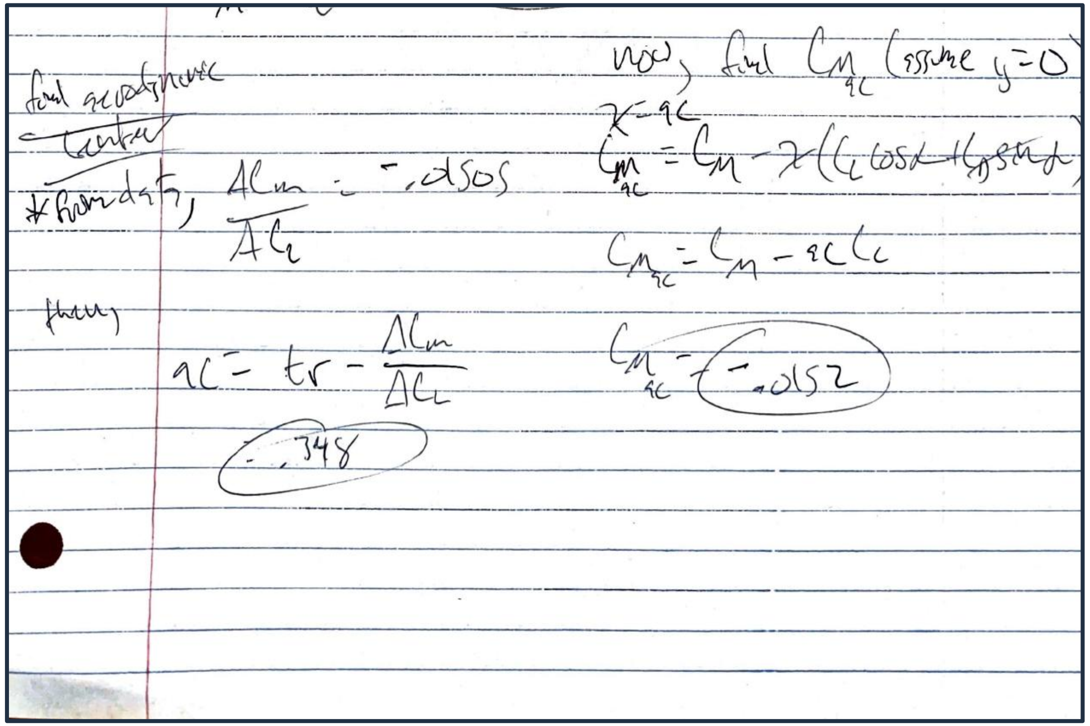


Figure [18]. - minor 3D printing errors

Sample Calculations





Figure(s) [19.1 - 19.2]. - Hand-calculation process done for C_L, C_D, and C_M

Discussion and Conclusion

Efficiency Improvements:

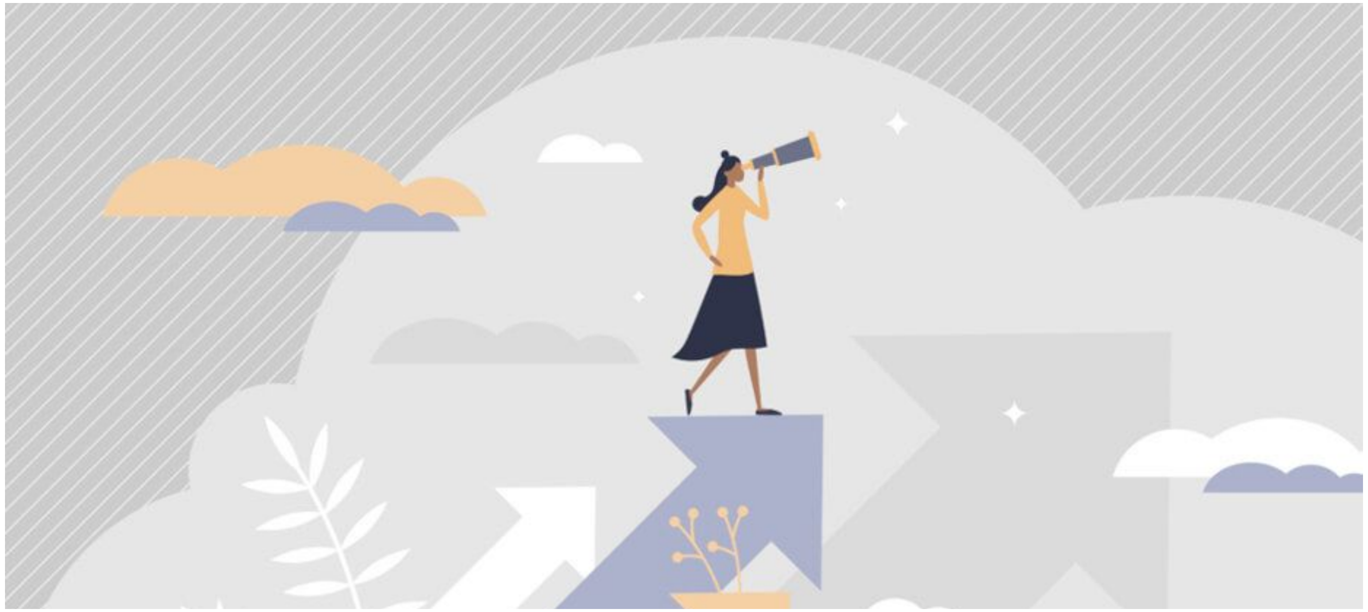
- Modern airfoils: Better lift-to-drag ratios.
- Wind tunnel tests: Theoretical predictions confirmed.

Performance Highlights:

- Historical airfoils: Higher drag, lower efficiency.
- Modern airfoils: Optimized shapes, better performance.

Suggestions for future repetitions of this experiment:

- Utilize more accurate, reliable, and capable equipment
- Compare to the performance of other historical airfoils
- Source all of the airfoil configurations from the same 3D printer



Thank you!

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